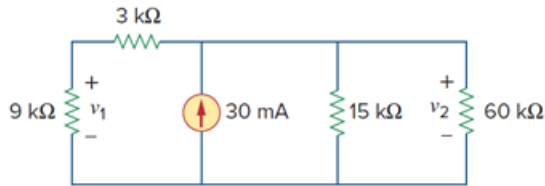


Problem

For the circuit shown in Fig. 2.45, find: (a) v_1 and v_2 , (b) the power dissipated in the 3-k Ω and 20-k Ω resistors, and (c) the power supplied by the current source.

Figure 2.45:



Step-by-step solution

Step 1 of 7

(a)

Refer to circuit diagram in Figure 2.45 in the textbook.

The circuit diagram with the current markings is as shown in Figure 1.

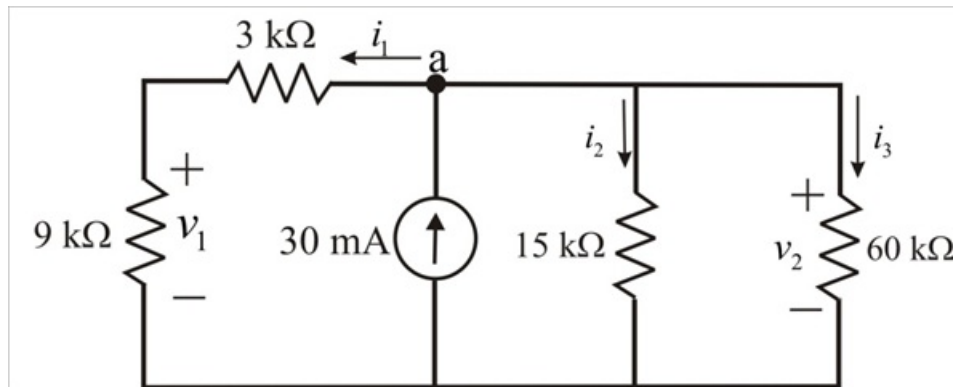


Figure 1

Step 2 of 7

The current i_1 is,

$$i_1 = \frac{v_2}{3 \text{ k}\Omega + 9 \text{ k}\Omega}$$

$$= \frac{v_2}{12 \text{ k}\Omega} \dots\dots (1)$$

Similarly, the currents i_2 and i_3 are,

$$i_2 = \frac{v_2}{15 \text{ k}\Omega} \dots\dots (2)$$

And,

$$i_3 = \frac{v_2}{60 \text{ k}\Omega} \dots\dots (3)$$

Step 3 of 7

Apply Kirchhoff's current law at node a.

$$\frac{v_2}{12 \text{ k}\Omega} + \frac{v_2}{15 \text{ k}\Omega} + \frac{v_2}{60 \text{ k}\Omega} = 30 \text{ mA}$$

$$\frac{5v_2 + 4v_2 + v_2}{60 \text{ k}\Omega} = 30 \text{ mA}$$

$$\frac{10v_2}{60 \text{ k}\Omega} = 30 \text{ mA}$$

$$v_2 = 180 \text{ V}$$

Therefore, the voltage, v_2 is 180 V.

Step 4 of 7

Recall equation (1).

$$i_1 = \frac{v_2}{12 \text{ k}\Omega}$$

Substitute 180 V for v_2 in the above equation.

$$\begin{aligned} i_1 &= \frac{180 \text{ V}}{12 \text{ k}\Omega} \\ &= 15 \text{ mA} \end{aligned}$$

Calculate the voltage v_1 .

$$\begin{aligned} v_1 &= (i_1)(9 \text{ k}\Omega) \\ &= (15 \text{ mA})(9 \text{ k}\Omega) \\ &= 135 \text{ V} \end{aligned}$$

Therefore, the voltage, v_1 is 135 V.

Step 5 of 7

(b)

Calculate the power dissipated in $3 \text{ k}\Omega$ resistor.

$$\begin{aligned} p_{3k} &= (i_1)^2 R \\ &= (15 \text{ mA})^2 (3 \text{ k}\Omega) \\ &= (225 \times 10^{-6})(3 \text{ k}\Omega) \\ &= 675 \text{ mW (Absorbing)} \end{aligned}$$

Therefore, the power dissipated in $3 \text{ k}\Omega$ resistor is 675 mW.

Step 6 of 7

Calculate the power dissipated in $60 \text{ k}\Omega$ resistor.

$$\begin{aligned} p_{60k} &= \frac{v_2^2}{60 \text{ k}\Omega} \\ &= \frac{(180 \text{ V})^2}{60 \text{ k}\Omega} \\ &= \frac{32400 \text{ V}^2}{60 \text{ k}\Omega} \\ &= 540 \text{ mW (Absorbing)} \end{aligned}$$

Therefore, the power dissipated in $60 \text{ k}\Omega$ resistor is 540 mW.

(c)

Calculate the power supplied by the current source.

$$\begin{aligned} p_{\text{source}} &= v_2 (30 \text{ mA}) \\ &= (180 \text{ V})(30 \text{ mA}) \\ &= 5.4 \text{ W (Delivering)} \end{aligned}$$

Therefore, the power supplied by the current source is 5.4 W.